

Quantitative Business Methods

Supplementary Teaching Material

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Review of Mathematics

0.1 Arithmetic Operations

Let us start with elementary arithmetic operations which we will need to use in this module. The acronym to remember is **BODMAS**, which tells you the correct order in which to perform a mathematical operation:

Operation	Example
Brackets	$()$
Orders	2^3
Division	$8 \div 2$
Multiplication	4×3
Addition	$5 + 2$
Subtraction	$7 - 3$

Example 1: Derive the equation:

$$(35 \div 7 + 10) - (4^2 - 6 \times 2)$$

Solution

Remember the order of **BODMAS**.

For convenience, let us call $(35 \div 7 + 10)$ the **first term** and $(4^2 - 6 \times 2)$ the **second term**. In order to subtract these two terms, you need to first evaluate the equations inside the **bracket**.

Lets start with the first term:

$$= 35 \div 7 + 10 \quad \text{Division}$$

$$= 5 + 10 \quad \text{Addition}$$

$$= 15$$

Now we can do the second term:

$$= 4^2 - 6 \times 2 \quad \text{Order}$$

$$= 16 - 6 \times 2 \quad \text{Multiplication}$$

$$= 16 - 12 \quad \text{Subtraction}$$

$$= 4$$

Finally, we can subtract the two terms:

$$= (35 \div 7 + 10) - (4^2 - 6 \times 2)$$

$$= 15 - 4$$

$$= 11$$

With practice, you will be able to conduct these operations with ease. It will be useful when we start with algebraic notations.

It is also worth remembering some interchanging notation. For *any* numbers a and b :

- The **sum** of a and b means $a + b$.
- The **difference**^a between a and b means $a - b$.
- The **product** of a and b means $a \times b = a \cdot b$.
- The **quotient** of a and b means a divided by b or a/b .

The *power* is the amount of times a number is multiplied by itself. For example:

$$3^4 = 3 \cdot 3 \cdot 3 \cdot 3 = 81$$

$$2^2 = 2 \cdot 2 = 4$$

One important power which we will use throughout the module is the square root $\sqrt{a}$. For example^b:

$$= \sqrt{9} = 3$$

$$= \sqrt{4} = 2$$

0.2 Fractions and Percentages

A fraction is simply a part of a whole and expressed as:

$$\frac{a}{b}$$

We call a the **numerator** while b is the **denominator**. You should be comfortable with handling fractions. Moreover,

^aThe reverse is also true, where $b - a$.

^bFor your information, the square root of a number is simply the base multiplied by 0.5.

you should also know that

$$\frac{a}{b} \cdot 100,$$

will convert the fraction into a percentage, with the % sign to be used.

Example 2:

(a) Evaluate this expression as a fraction:

$$\frac{4}{9} \cdot \frac{1}{3} \cdot \frac{2}{5}$$

(b) Express the fraction as a percentage.

Solution

(a) You can simply multiply all the numerator terms together and all the denominator terms together:

$$\frac{4 \cdot 1 \cdot 2}{9 \cdot 3 \cdot 5} = \frac{8}{135}$$

(b) To convert this into a percentage, simply multiply it by 100.

$$\frac{8}{135} \cdot 100 = 5.92\%$$

0.3 Functions

You should be very comfortable in understanding what a function is and how to illustrate it.

When a variable y depends on another variable x , we can mathematically denote this as:

$$y = f(x)$$

where we say that y is a function of x .

Example 3: Consider the below function:

$$y = 2x$$

(a) Express each value of y for the below values of x :

x	y
0	
1	
2	
3	
4	
5	

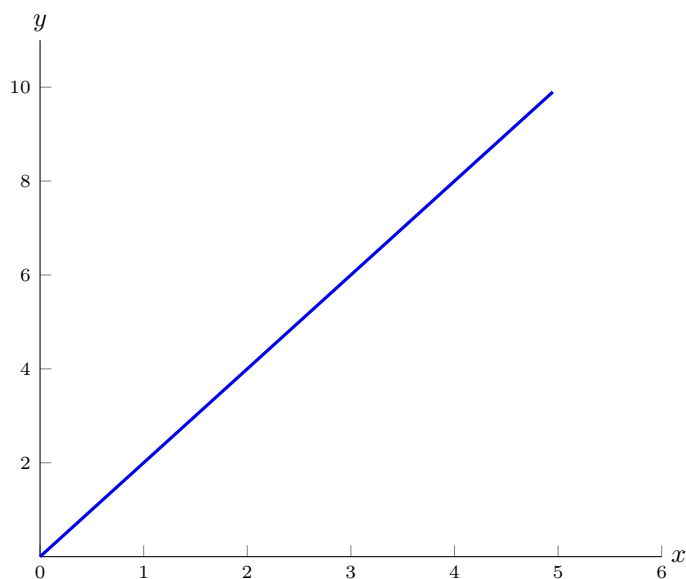
(b) Plot a graph where y is on the vertical axis and x is on the horizontal axis. Label each value of (x, y) and connect it.

Solution

(a) Simply input each value of x into the function $y = 2x$:

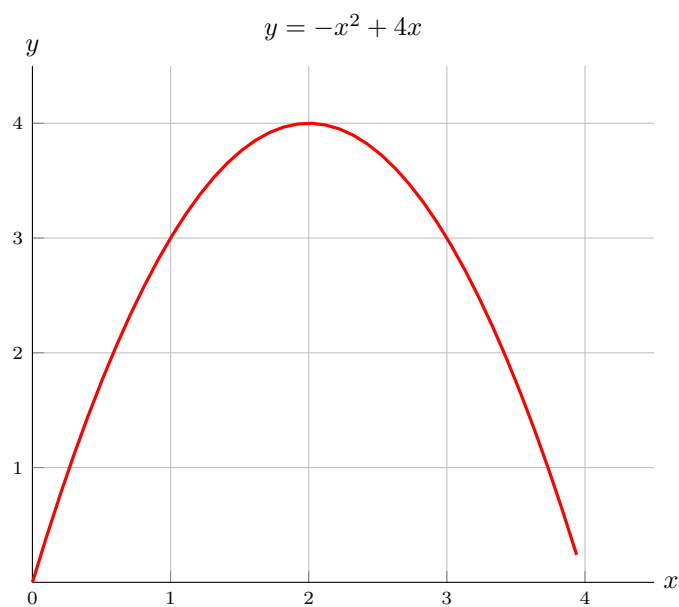
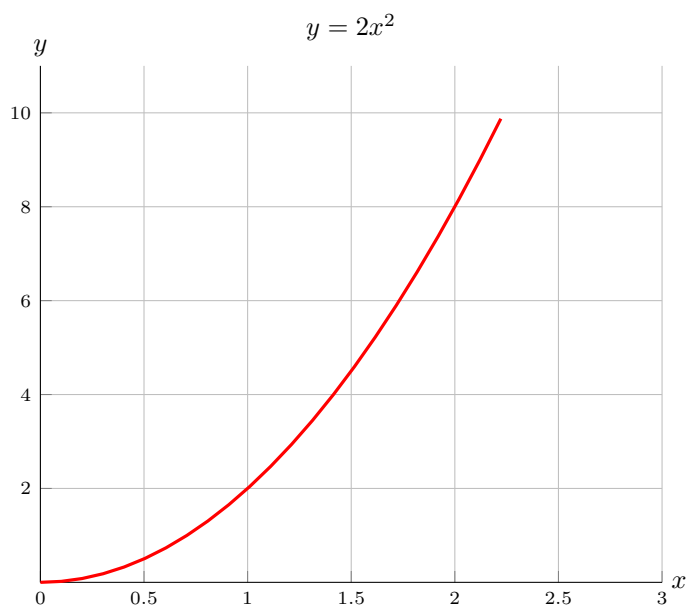
x	y
0	0
1	2
2	4
3	6
4	8
5	10

(b) As per the instructions, we can illustrate this function as below:



Notice how the function is a straight line. We call this a **linear function**^c.

A non-linear function can be visually identified as having curves^d:



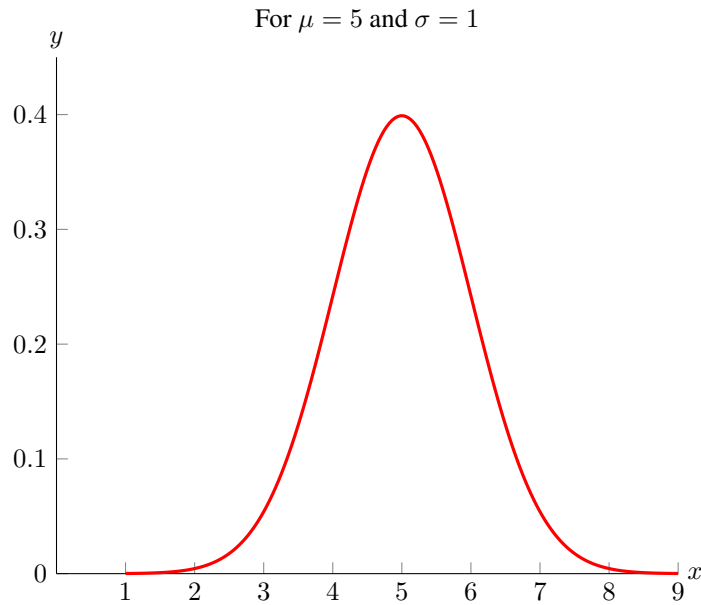
An important non-linear function that you will need to be familiar with is the **normal distribution**:

^cThe qualifier here is that the slope of a linear function is constant. You can derive this using a variety of methods, but one way is through calculus. For **Example 3** the slope of the function is $\frac{dy}{dx} = 2$.

^dSimilarly, a non-linear function will not have constant slopes. In fact, the slope changes based on the position of x .

$$y = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{\sigma^2}}$$

Although this function looks complicated^e it is actually quite simple to calculate in a software if you have the values of the mean μ and the variance σ^2 . You will learn more about this later in the module. For now, you should just know that it is a non-linear function and can be visually represented as below:



0.4 Summation Operator

Consider the exam score data of 5 students from a class:

i	Name	Score x
1	Ahmed	70
2	Sara	90
3	Hassan	80
4	Fatima	44
5	Zahra	73
6	Hisham	32

^eThe normal distribution function describes the probability of observing a particular value of x . It says that it is greatest when it is near the average μ , and that this probability dies away rapidly as the difference from the mean increases. The rapidity of the degeneration is based on the standard deviation σ .

There are a total of $n = 6$ people in this dataset. Moreover, each person is indexed by a number i . For example, x_5 refers to Zahra's score, which is $x_5 = 73$.

Suppose we are interested in the sum of all scores in the class. This is simply:

$$= 70 + 90 + 80 + 44 + 73 + 32 = 389$$

Another way to express the sum of the score variable x is by using the summation operator $\sum$. Formally, we would say:

$$= \sum_{i=1}^n x_i$$

$$= x_1 + x_2 + x_3 + x_4 + x_5 + x_6$$

$$= 70 + 90 + 80 + 44 + 73 + 32$$

$$= 389$$

This will be a useful operator later in the module. For example, the average score of the class can be expressed as:

$$= \frac{1}{n} \sum_{i=1}^n x_i$$

$$= \frac{389}{6} = 64.8$$

0.5 Problem Set

1) Work out the following:

(a) $(2 + 4) \times (3 + 7)$

(b) $(1/3 \times 12) - 4/2$

(c) $\frac{(1+4)}{5} \times (100 - 98)$

2) Work out the following (You can use a calculator when needed):

(a) $\sqrt{16}$

(b) $(0.07)^2$

(c) $\sqrt{0.49}$

3) Work out the following:

(a) What is 98% of 200?

(b) Express $17/25$ as a percentage.

4) Sketch the following functions:

(a) $y = x + 3$

(b) $y = 3x - 2$

5) Consider the following data:

i	x
1	3
2	1
3	4
4	6
5	8

Find the following expressions:

(a) $\sum_{i=1}^5 x_i$.

(b) $\frac{1}{n} \sum_{i=1}^5 x_i^2$.

0.6 Solutions

1) Remember to use BODMAS:

(a)

$$\begin{aligned}(2 + 4) \times (3 + 7) &= 6 \times 10 \\ &= 60\end{aligned}$$

(b)

$$\begin{aligned}\left(\frac{1}{3} \times 12\right) - \frac{4}{2} &= 4 - 2 \\ &= 2\end{aligned}$$

(c)

$$\begin{aligned}\frac{(1 + 4)}{5} \times (100 - 98) &= \frac{5}{5} \times 2 \\ &= 1 \times 2 \\ &= 2\end{aligned}$$

2) You can use a calculator for the second and third part:

(a)

$$\sqrt{16} = 4$$

(b)

$$(0.07)^2 = 0.0049$$

(c)

$$\sqrt{0.49} = 0.7$$

3) Remember that a percentage is a fraction multiplied by 100:

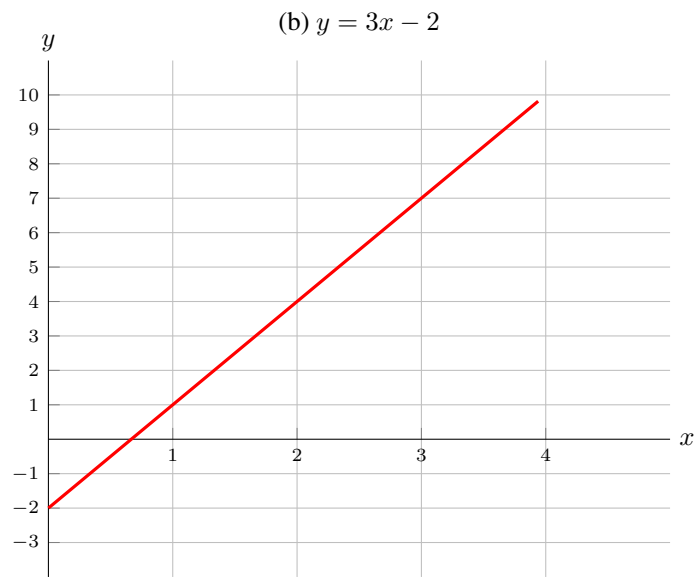
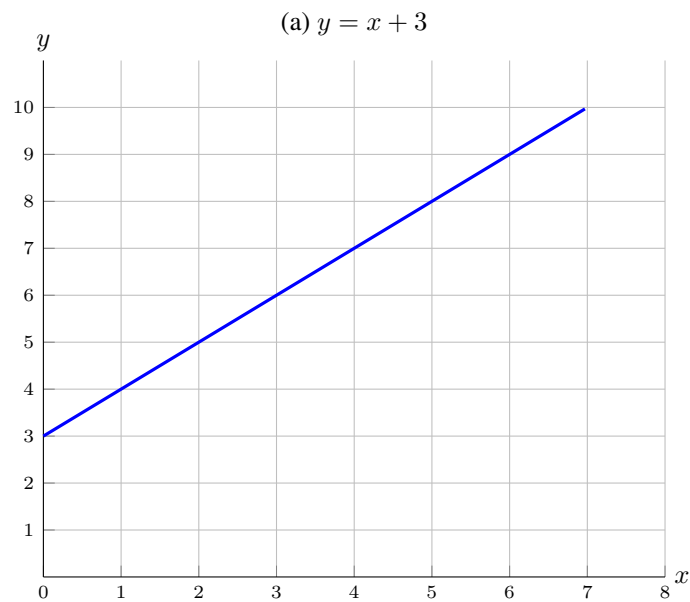
(a)

$$\begin{aligned} 98\% \text{ of } 200 &= \frac{98}{100} \times 200 \\ &= 196 \end{aligned}$$

(b)

$$\frac{17}{25} \times 100 = 68\%$$

4) Note that both functions are linear. However for (b), the intercept point starts at a negative number.



5) Using the summation operator:

(a)

$$\sum_{i=1}^5 x_i = 3 + 1 + 4 + 6 + 8$$

$$= 22$$

(b)

$$\frac{1}{n} \sum_{i=1}^5 x_i^2 = \frac{1}{5} (3^2 + 1^2 + 4^2 + 6^2 + 8^2)$$

$$= \frac{1}{5} (9 + 1 + 16 + 36 + 64)$$

$$= \frac{126}{5}$$

$$= 25.2$$

Chapter 1

What is Statistics

1.1 Descriptive and Inferential Statistics

You are a policy analyst of the Bahraini Ministry of Education and you have been asked to do a study on the quality of primary schools in Bahrain. You collected data on a sample of schools and inputted it into a dataset:

i	School	Type	Score
1	Ain Jaloot Primary Intermediate Girls	Female	Good
2	Abusaiba Primary Boys	Male	Bad
3	Aisha Um Al Moamneen Primary Girls	Female	Good
4	A'ali Primary Boys	Male	Good
5	Sanad Primary Girls	Female	Bad
6	Ahmed Al Fateh Primary Intermediate Boys	Male	Bad
7	Al-Budaiyya Primary Intermediate Girls	Female	Good
8	Hajer Primary Girls	Female	Bad
9	Al Imam Malek Bin Anas Primary Boys	Male	Bad
10	Al Busaiteen Primary Girls	Female	Good
11	Abu Bakr Al-Siddeeq Primary Intermediate Boys	Male	Bad
12	Al Maamoon Primary Boys	Male	Bad
13	Al-Belad Al-Qadeem Primary Girls	Female	Good
14	Al Andalus Primary Girls	Female	Good
15	East Riffa Primary Boys	Male	Good
16	Abu Alaala Almaari Primary Boys	Male	Good
17	Al-Alaa Al Hadrami Primary Boys	Male	Good
18	A'ali Primary Girls	Female	Good
19	Al Budaiya Primary Boys	Male	Good
20	Al Malikiyah Primary Intermediate Girls	Female	Good

In order to conduct your study, you summarise the information in a small table:

	Good	Bad
Boys	5	5
Girls	8	2

There are a number facts that you can use to describe your data:

- There are a total of 20 schools in your dataset.
- There are 10 boy schools in your sample.
- There are 10 girl schools in your sample.
- 50% of boys schools achieved a good score.
- 80% of girls schools achieved a good score.

This is known as **descriptive statistics**. You are simply looking at the information that was collected and describing the contents.

Now suppose you are interested to find out whether girl schools are *better* than boys schools. One primitive way to doing this is to simply compare the percentage difference of good schools based on gender. We can clearly see that

$$80\% > 50\%,$$

Based on the sample, an analyst can conclude that **all** primary girl schools in Bahrain are higher quality than **all** boy schools.

This analysis is known as **inferential statistics**. You have analysed your sampled data and drawn a conclusion on all schools in Bahrain.

Activity 1:

You currently work as a strategist for Batelco. Explain whether the following tasks are considered **descriptive** or **inferential** statistics:

- (a) The average network engineer works for 1.5 years before leaving the company.

- (b) Based on a sample of customers, 68% said they would not switch their service, which means that most people are happy with the service.

Solution

- (a) The qualifies as **descriptive** statistics, because we are simply talking about the data collected from network engineers.
- (b) The qualifies as **inferential** statistics, since you are analysing the response from a sample of customers.

1.2 Sample and Population

An important fact to consider is that your analysis is only based on 20 schools. Suppose that three other Ministry of Education employees also did their own investigations and collected the following data from 20 schools each:

Sample X			Sample Y			Sample Z		
	Good	Bad		Good	Bad		Good	Bad
Boys	4	6	Boys	5	5	Boys	6	4
Girls	9	1	Girls	7	3	Girls	8	2

Even though they all have different numbers, the overall story remains the same. It seems that based on the sample of schools, each investigator can make an inference that primary girls schools are better than primary boys schools.

These are known as **samples**. They represent parts of the overall total of primary schools across Bahrain.

Suppose you now go and collect data on **all** 130 primary schools in Bahrain. You find the following information:

	Good	Bad
Boys	53%	47%
Girls	96%	4%

Since you have collected this information from all 130 schools, then we say that this dataset is the **population** of primary schools in Bahrain.

An important point to keep in mind is that inferential statistics allows us to draw a conclusion about the population based on a sample.

1.3 Types of Variables

In order to assess whether a public school is good or bad, you need to base your conclusion a set of variables. For example:

- Funding received from the government.
- Number of teachers and students.
- Number of classrooms
- Classroom size.
- Exam scores.
- Number of air conditioners.

These are examples of quantitative variables. That is, the data that you record will be inputted as numbers.

School	Funding (BD)	Students	Teachers	Classrooms	Class Size	Avg. Exam Score (%)	A/Cs
School 1	450,000	600	45	25	24	78	32
School 2	520,000	750	50	30	25	84	38
School 3	380,000	480	35	20	24	71	25
School 4	600,000	800	55	32	25	89	40

You should be aware that there are two types of quantitative variables: **continuous** and **discrete**.

Continuous variables can be described as numbers that can value within a range. For example, the amount of money received by a school in 2026 can 2.6 Million BD or the average score of the mathematics exam was 67.3%.

Discrete variables can be described as those that can counted. For example, a school can be 50 teachers, 100 classrooms and each classroom can have 2 ACs. A school **cannot** have 100.5 classrooms or 50.3 teachers or 0.5 ACs. This is what makes it a discrete variable.

Data can also be **qualitative**. This means that the data collected is some descriptive measure and not something measurable. For example:

- City where the school is located.
- The education degree of the principal.
- If there is a requirement to wear uniforms.

School	City	Principal's Degree	Uniform Required
School 1	Manama	Bachelor's	Yes
School 2	Riffa	Master's	Yes
School 3	Muharraq	PhD	No
School 4	Hamad Town	Master's	Yes

1.4 Levels of Measurement

Both quantitative and qualitative data can be categorised by four levels of measurement:

Level of Measurement	Description	Examples
Nominal	Data may only be classified into different categories.	Employee department; Firm's industry; School type
Ordinal	Data are classified into categories that can be ranked or ordered.	Customer satisfaction; Credit rating; Performance in a Class
Interval	Data are ordered and the differences between values are meaningful.	Temperature in degrees Celsius; Calendar years
Ratio	Data have meaningful differences between values and a meaningful zero point.	Employee salary; Sales; Number of employees; School enrolment

One way to identify the difference between interval and ratio is to ask if *is zero a genuine absence of the measured quantity?*.

Only ratio requires zero to be a proper "zero". For example, 0 BD, 0 employees, 0 students in a classroom, where the zero is properly understood. However, 0°C does not literally mean "no temperature" and "Year 0" does not literally mean "no years".

1.5 Ethics in Statistics

We conclude this chapter by informing students that statistics can easily be manipulated or misrepresented. As a statistician, you must try to collect data in principled way.

- Make sure you collect a proper representative sample.
- Do not omit relevant statistics.
- Do not let bias impact your collection.

Activity 3

Explain how each scenario impacts the school study.

- (a) You only collect data on schools from high-income neighbourhoods.
- (b) You notice that teachers in boy's schools get paid much less compared to girl's schools, but cannot collect the data.
- (c) The principal only lets you interview teachers with high evaluations.

Solution:

- (a) By omitting schools from low-income areas, your sample is not representative of the population. The average scores, performance and ranks of the schools will be higher than what is true.
- (b) Its possible that because teachers in boy's schools get paid less, they are not as good as teachers in girl's schools. This is probably useful information that can explain the difference in performance that you would report to the ministry of education.
- (c) Your data will be biased because you are only getting information from teachers that perform well in the school. Low-performing teachers will not be considered and so you will improperly rank these schools.

1.6 Problem Set

1) For each of the following statements, identify whether it is an example of **descriptive statistics** or **inferential statistics**.

- (a) A company records the salaries of all 50 of its employees and reports that the average salary is BD 850 per month.
- (b) A supermarket surveys 100 customers and uses the results to estimate the percentage of all its customers who are satisfied with its service.
- (c) The monthly sales of a business are recorded for the past 12 months and presented using a line graph.
- (d) A researcher surveys 500 households in Bahrain and uses the results to estimate the average household income of all households in Bahrain.
- (e) A university reports that 72% of the 200 students who completed a survey were satisfied with their course.
- (f) A shop records the amount spent by 150 customers on a particular day and calculates a average expenditure of BD 12.
- (g) A researcher examines a sample of 100 firms and concludes that larger firms tend to employ more workers than smaller firms in the wider population of businesses.

2) Consider the below cases and specify the **type** (qualitative or quantitative) and **level** (nominal, ordinal, interval, ratio) for each measure:

- (a) Eye Colour.
- (b) Rank in a class room.
- (c) Temperature.
- (d) Annual Salary.

Specify an example of a data point for each case.

1.7 Solutions

1)

- (a) A company records the salaries of all 50 of its employees and reports that the average salary is BD 850 per month. **DESCRIPTIVE.**
- (b) A supermarket surveys 100 customers and uses the results to estimate the percentage of all its customers who are satisfied with its service. **INFERENTIAL.**
- (c) The monthly sales of a business are recorded for the past 12 months and presented using a line graph. **DESCRIPTIVE.**
- (d) A researcher surveys 500 households in Bahrain and uses the results to estimate the average household income of all households in Bahrain. **INFERENTIAL.**
- (e) A university reports that 72% of the 200 students who completed a survey were satisfied with their course. **DESCRIPTIVE.**
- (f) A shop records the amount spent by 150 customers on a particular day and calculates a average expenditure of BD 12. **DESCRIPTIVE.**
- (g) A researcher examines a sample of 100 firms and concludes that larger firms tend to employ more workers than smaller firms in the wider population of businesses. **INFERENTIAL.**

2)

- (a) Eye colour: Brown, blue, green.

Eye colour is a **qualitative** variable because its values represent categories rather than numerical quantities. It has a **nominal** level of measurement because there is no natural ordering of the categories. For example, brown cannot meaningfully be ranked above or below blue or green.

- (b) Class Rank: 1st place, 2nd place,...

Class rank is a **qualitative** variable with an **ordinal** level of measurement. Although numbers are used to represent the ranks, these numbers indicate positions rather than numerical quantities. We can place the observations in a meaningful order: a student ranked 1st is ahead of a student ranked 2nd, who is ahead of a student ranked 3rd.

However, the differences between ranks are not meaningful numerical distances. For example, the difference between 1st and 2nd place is not necessarily equivalent to the difference in performance between 2nd and 3rd place.

(c) Temperature: 18° C

Temperature measured in degrees Celsius is a **quantitative** variable because its values represent measurable numerical quantities. It has an **interval** level of measurement because differences between values are meaningful. For example, the difference between 10°C and 20°C is the same temperature difference as between 20°C and 30°C.

However, Celsius does not have a meaningful zero point. A temperature of 0°C does not represent the absence of temperature. Consequently, ratios are not meaningful: we cannot say that 20°C is “twice as hot” as 10°C.

(d) Annual Salary: 1000 BD

Annual salary is a **quantitative** variable because it represents a measurable numerical quantity. It has a **ratio** level of measurement because differences between values are meaningful and there is a meaningful zero point. An annual salary of BD 0 represents no salary income.

Ratios are therefore also meaningful. For example, an annual salary of BD 20,000 is twice an annual salary of BD 10,000.

Chapter 2

Describing Data: Frequency Tables and Graphs

In this chapter, we are interested in the various ways in which we can present qualitative and quantitative data.

2.1 Qualitative Data

2.1.1 Frequency Table

The Bahraini Minister of Urban Planning has received a complaint that petrol station queue times are very long. The minister asks you to collect data on all petrol stations in order to conduct a study.

After collecting 180 entries, your dataset is shown as below:

i	Petrol Station Name	Town	Governorate
1	Al Fateh Service Station	Manama	Capital
2	Bapco Energies Station – Muharraq	Muharraq	Muharraq
3	Bapco Energies Station – Sanabis	Sanabis	Capital
4	Salmabad Service Station	Salmabad	Northern
5	Galali Petrol Station	Galali	Muharraq
6	Al Buhair Petrol Station	Riffa	Southern
$\vdots$			
180	Hajjyat Petrol Station 180	Riffa	Southern

There are various ways to present this data to the minister. One way is to construct a frequency table.

A **Frequency Table** is a grouping of qualitative data into mutually exclusive and collectively exhaustive classes showing the number of observations in each class.

Mutually Exhaustive implies that each observation falls on only one class.

Collectively Exhaustive implies that there is a class for every value.

An *observation* refers to a row in your dataset. For example, East Riffa Petrol Station is one observation in your dataset.

Let us start with a basic analysis and find out how many petrol stations are available in each governorate:

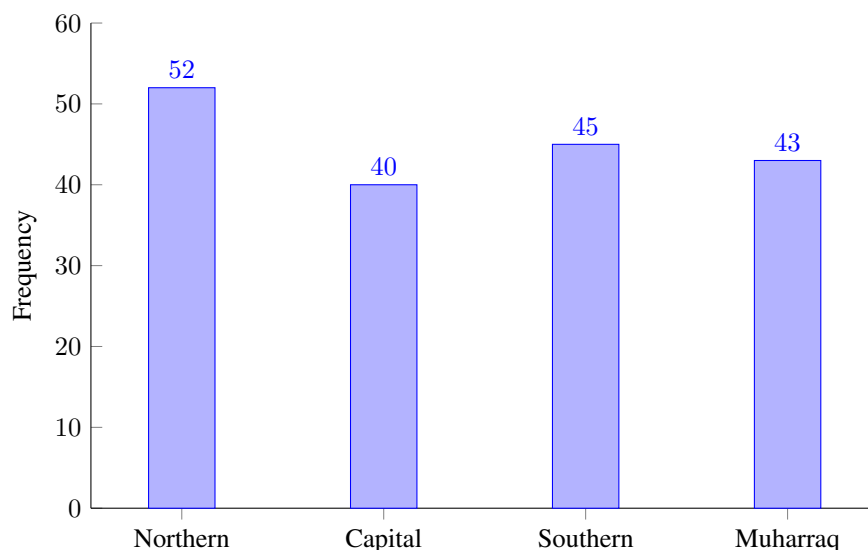
Governorate	Frequency
Northern	52
Capital	40
Southern	45
Muharraq	43
Total	180

This is a simple way of presenting this information to the Minister. Notice that there are 4 governorates: Northern, Capital, Southern and Muharraq. For the frequency table, we call each of these categories as **classes**. We call this a frequency table because it shows the total amount of **observations** for each **class**. For example, there are 40 petrol stations in the Capital governorate.

2.1.2 Bar Chart

You can take your dataset on petrol stations and display it in a **bar chart**. This is when we have the frequency of the data on the vertical axis while each class is represented by a bar on the horizontal axis.

Figure 2.1: Frequency of Petrol Stations by Governorate



The bar chart is informative because it immediately reveals the class with the highest frequency.

We will learn later that the **mode** of this dataset is the class which has the highest frequency. In this case, it is the capital governorate.

2.1.3 Pie Chart

It is probably more informative to display this information with the **share** of each class to the overall total. To do this, we simply divide the number of petrol stations from each class with the total. We call this the **relative frequency**.

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}}$$

Since there are a total of 180 petrol stations, we can arrive at the following result:

Governorate	Frequency	Calculation	Relative Frequency
Northern	52	$\frac{52}{180}$	0.289
Capital	40	$\frac{40}{180}$	0.222
Southern	45	$\frac{45}{180}$	0.250
Muharraq	43	$\frac{43}{180}$	0.239
Total	180	$\frac{180}{180}$	1.000

You can use this information to construct a **pie chart**. This displays the relative frequency of each class.

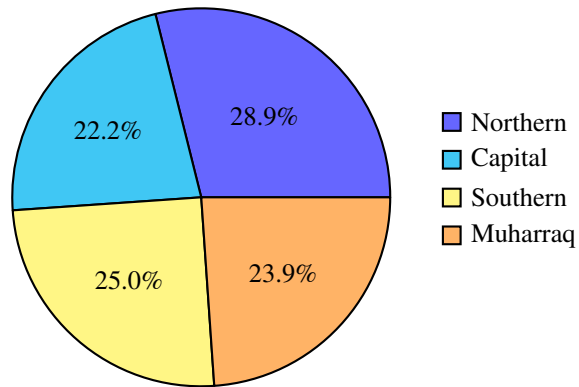


Figure 2.2: Petrol Stations by Governorate

Here, each *slice* of the chart represents a class. For example, 22.2% of the petrol stations in your dataset is from the Capital governorate.

Activity 1:

Bahrain City Center has the following aggregated information about types of stores in their mall:

Store Type	Frequency
Clothing and Fashion	70
Food and Beverage	45
Electronics	20
Health and Beauty	30
Home and Furniture	15
Entertainment and Leisure	20
Total	200

- Create an additional column representing the relative frequency.
- Display the totals of each class in a bar chart.
- Display the relative frequency of each class in a pie chart.

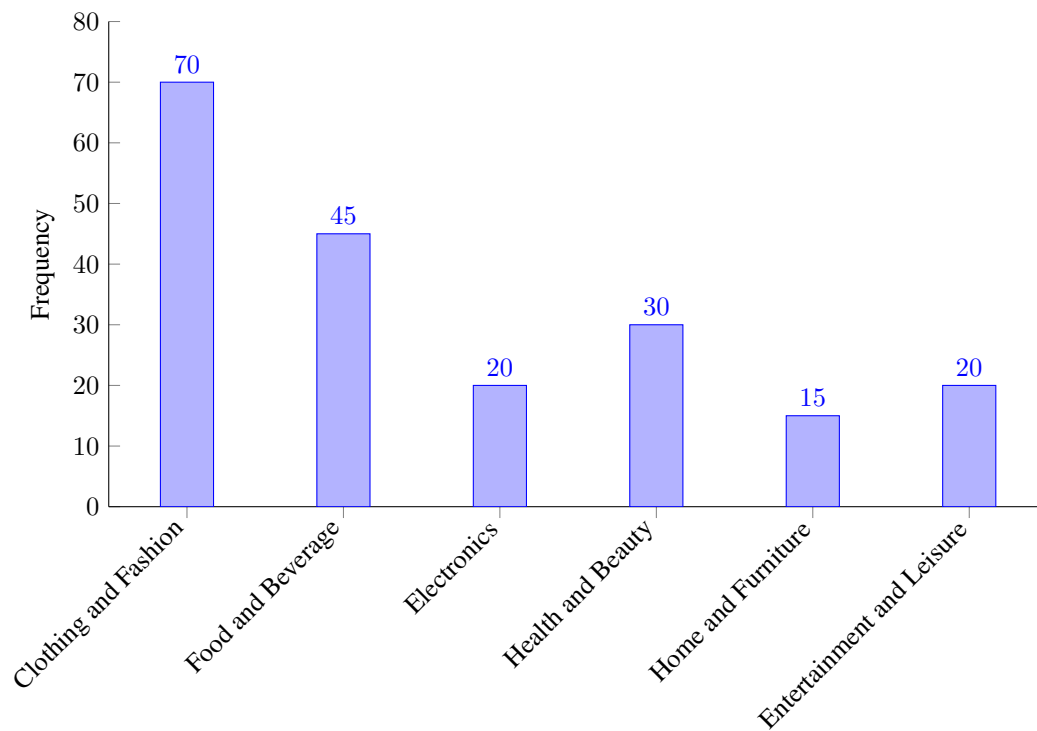
Solution:

- To create the relative frequency column, we first get the total number of store in City Center. we then divide the frequency of each class with the total number of stores.

Store Type	Frequency	Relative Frequency
Clothing and Fashion	70	0.350
Food and Beverage	45	0.225
Electronics	20	0.100
Health and Beauty	30	0.150
Home and Furniture	15	0.075
Entertainment and Leisure	20	0.100
Total	200	1.000

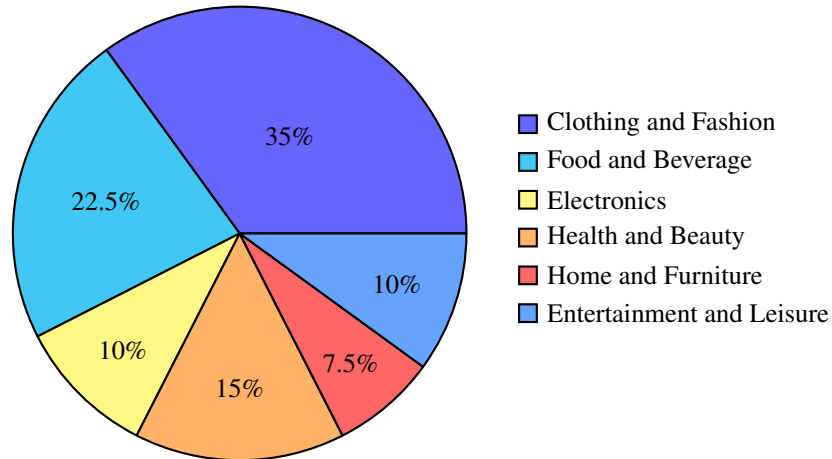
(b) We can display the totals in a bar chart:

Figure 2.3: Frequency of Stores by Type



(c) We can display the relative frequency in a pie chart:

Figure 2.4: Relative Frequency of Stores by Type



2.2 Quantitative Data

Previously, we worked with displaying qualitative data as classes in a frequency table. This means that each class was some qualitative category, like the governorate or the type of store in a mall. In this section, we will study a similar analysis, except this time the classes will be quantitative data.

2.2.1 Frequency Table

Consider a dataset on the sale of cars for the Toyota Showroom in September 2026:

Transaction Number	Sales Value (BD)
1	15,520
2	23,590
3	19,190
4	27,530
5	24,360
6	20,310
⋮	
180	30,380

Sales data can have many observations. It would be useful to create a frequency table to properly summarise this information. Just as before, we need to create a set of groups (classes). However, since we only have quantitative data, we would need to create the class based on an **interval of values**.

Suppose we created the frequency of transactions under these classes:

- 2,000 up to 12,000
- 12,000 up to 24,000
- 24,000 up to 34,000

That is, there are three classes, and each one has a width of 10,000 BD:

Interval Class (BD)	Frequency
2,000 up to 12,000	5
12,000 up to 24,000	75
24,000 up to 34,000	100
Total	180

This is how we would display a frequency table with quantitative data classes. There are 5 transactions that are valued in the first class, 75 transactions in the second class, and 100 transactions in the third class.

We arbitrarily selected very few classes to represent the dataset. Depending on the context of your study, you can also have many classes:

Interval Class (BD)	Frequency
2,000 up to 4,500	1
4,500 up to 7,000	0
7,000 up to 9,500	1
9,500 up to 12,000	3
12,000 up to 14,500	5
14,500 up to 17,000	10
17,000 up to 19,500	13
19,500 up to 22,000	22
22,000 up to 24,500	35
24,500 up to 27,000	24
27,000 up to 29,500	31
29,500 up to 32,000	29
32,000 up to 34,500	6
34,500 up to 37,000	0
Total	180

Now this shows that there are 14 classes, where each interval has a width of 1500 BD.

You should know that when deciding on the number of classes and the choice of interval width, we can follow a rule of thumb to construct the optimal amounts.

2.2.2 Constructing the Optimal Frequency Distribution

You need to decide on **the number of classes** and the **width of the interval** for each class. In order to do this, you need three details about your dataset:

- The total number of observations n .
- The minimum value of your variable in the dataset min.
- The maximum value of your variable in the dataset max.

The Number of Classes

Use the following rule:

$$2^k > n$$

where k is the number of classes and n is the total observations in your dataset.

Keep choosing different values of k until the equality is satisfied.

Width of the Interval

To determine the optimal interval, we use the following rule:

$$i = \frac{\max - \min}{k}$$

max and min are the largest and smallest values of the variable in your dataset. We normally round i to the nearest whole number or a convenient multiple of 10, 100, 1000, etc.

Activity 2: Consider the sales data regarding Toyota's showroom. We are given the following details:

- $n = 180$
- $\min = 2,940$ BD
- $\max = 32,920$ BD

Calculate the optimal number of classes k and interval i using the rules described earlier.

Solution:

For the number of classes, we need to satisfy the rule:

$$2^k > n$$

Since there are $n = 180$ observations, we can test for values of k until the equality is just satisfied.

For example, if $k = 7$:

$$2^7 = 128 < 180$$

which does not satisfy the condition. However if we do $k = 8$:

$$2^8 = 256 > 180$$

Therefore, $k = 8$ is the recommended number of classes.

As for the interval width, we need to simply compute:

$$i = \frac{\max - \min}{k}$$

$$i = \frac{39290 - 2940}{8} = 4543.75$$

We can round this to the nearest whole number to get to 4,500.

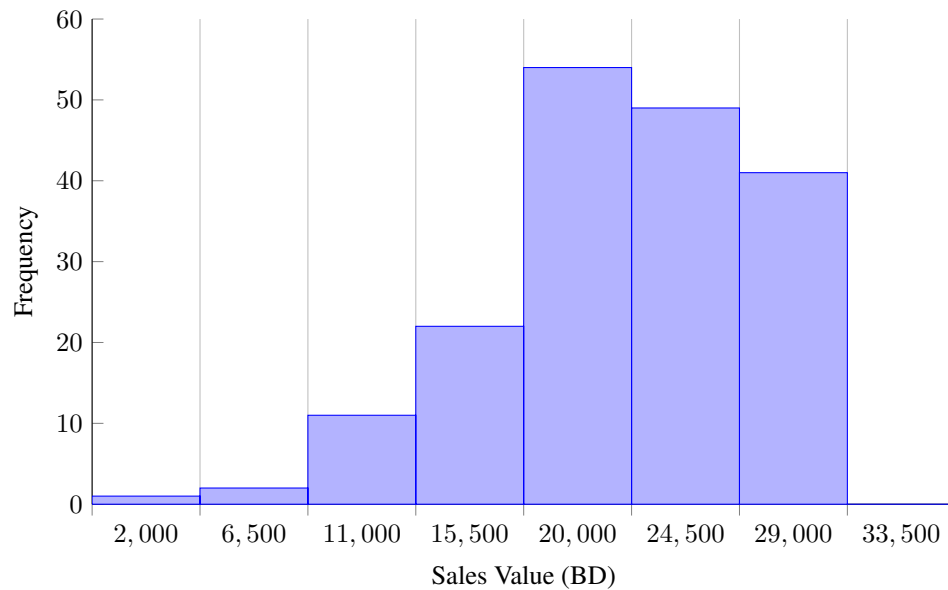
Now that we know how to calculate the optimal number of classes and interval width, we can display this information in a frequency table:

Interval Class (BD)	Frequency
2,000 up to 6,500	1
6,500 up to 11,000	2
11,000 up to 15,500	11
15,500 up to 20,000	22
20,000 up to 24,500	54
24,500 up to 29,000	49
29,000 up to 33,500	41
33,500 up to 38,000	0
Total	180

2.2.3 Histogram

We can take our optimised frequency table and display it in a **histogram**. This is very similar to a **bar chart**, where the vertical axis is the frequency and the horizontal axis represents the class intervals:

Figure 2.5: Distribution of Car Sales Values



2.2.4 Frequency Polygon

Another way to display quantitative data is through a **frequency polygon**. This is when the vertical axis has the frequency, but the horizontal axis is represented by the **midpoint** of every interval.

Midpoint of an Interval

To calculate the midpoint, simply apply the below formula for each interval class:

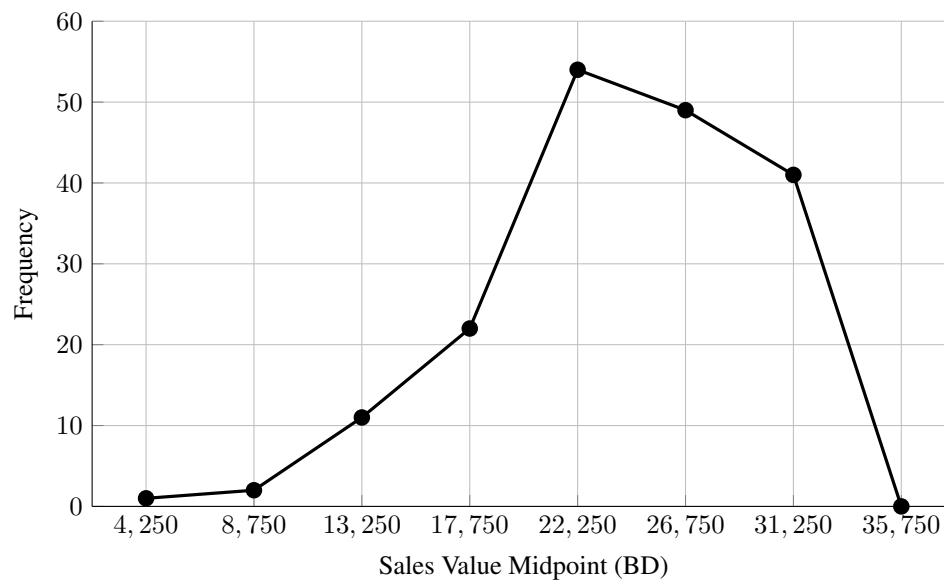
$$\frac{\text{Lower Bound} + \text{Upper Bound}}{2}$$

Using the car sale data:

Interval Class (BD)	Midpoint (BD)	Frequency
2,000 up to 6,500	4,250	1
6,500 up to 11,000	8,750	2
11,000 up to 15,500	13,250	11
15,500 up to 20,000	17,750	22
20,000 up to 24,500	22,250	54
24,500 up to 29,000	26,750	49
29,000 up to 33,500	31,250	41
33,500 up to 38,000	35,750	0

The frequency polygon can therefore be plotted:

Figure 2.6: Frequency Polygon of Car Sales Values



2.2.5 Relative Frequency

Similar to the previous section, we can also construct relative frequency distributions. Remember that the way to create this is take the total count of each class and divide it by the total number across all classes:

$$\text{Relative Frequency} = \frac{\text{Frequency}}{\text{Total Frequency}}$$

Interval Class (BD)	Frequency	Calculation	Relative Frequency
2,000 up to 6,500	1	$\frac{1}{180}$	0.006
6,500 up to 11,000	2	$\frac{2}{180}$	0.011
11,000 up to 15,500	11	$\frac{11}{180}$	0.061
15,500 up to 20,000	22	$\frac{22}{180}$	0.122
20,000 up to 24,500	54	$\frac{54}{180}$	0.300
24,500 up to 29,000	49	$\frac{49}{180}$	0.272
29,000 up to 33,500	41	$\frac{41}{180}$	0.228
33,500 up to 38,000	0	$\frac{0}{180}$	0.000
Total	180	$\frac{180}{180}$	1.000

2.2.6 Cumulative Frequency

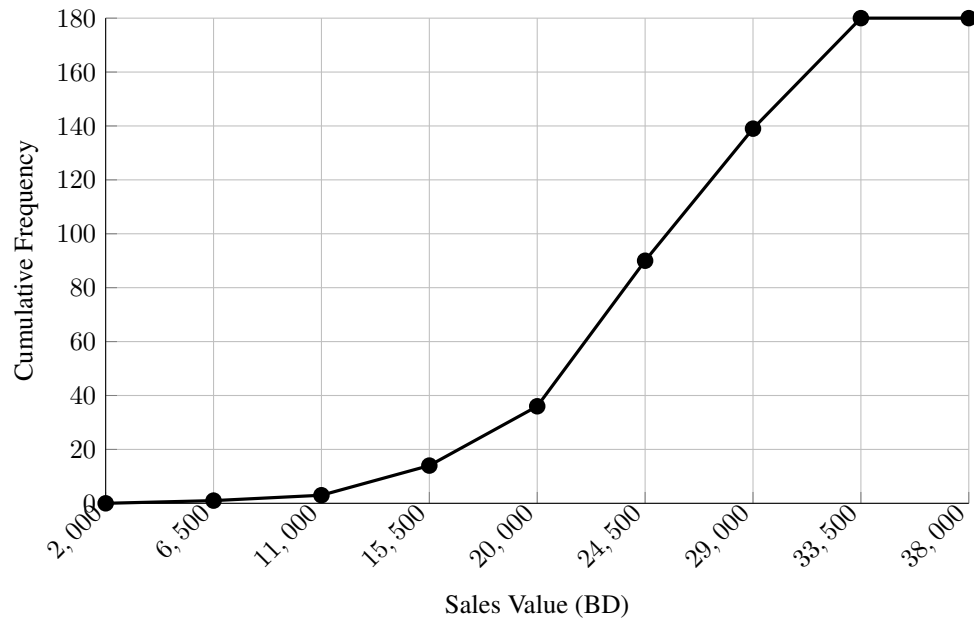
A cumulative frequency graph is one way of illustrating quantitative data across classes. Consider again the frequency graph for Toyota car sales:

Interval Class (BD)	Frequency
2,000 up to 6,500	1
6,500 up to 11,000	2
11,000 up to 15,500	11
15,500 up to 20,000	22
20,000 up to 24,500	54
24,500 up to 29,000	49
29,000 up to 33,500	41
33,500 up to 38,000	0
Total	180

The cumulative frequency is simply the addition of each row:

Interval Class (BD)	Frequency	Calculation	Cumulative Frequency
> 6,500	1	1	1
> 11,000	2	1 + 2	3
> 15,500	11	3 + 11	14
> 20,000	22	14 + 22	36
> 24,500	54	36 + 54	90
> 29,000	49	90 + 49	139
> 33,500	41	139 + 41	180

We can display this in a graph where the cumulative frequency is on the vertical axis and the class in on the horizontal axis:



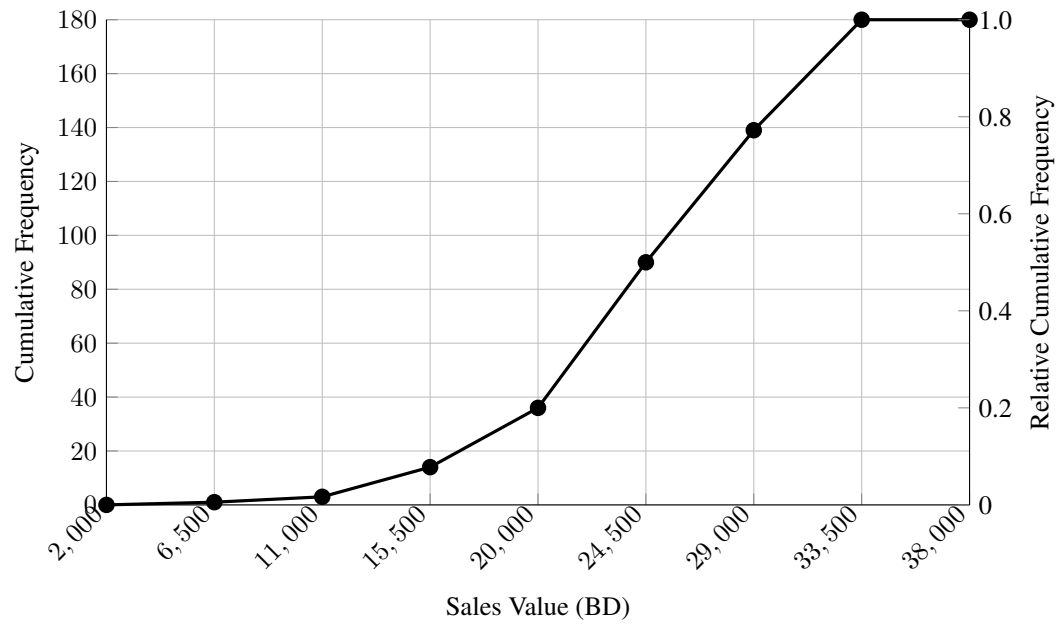
You can also gather the cumulative relative frequency to asses the contribution of each class.

$$\text{Relative Cumulative Frequency} = \frac{\text{Cumulative Frequency}}{\text{Total Frequency}}$$

Interval (BD)	Cumulative Frequency	Calculation	Relative Cumulative Frequency
< 6,500	1	$\frac{1}{180}$	0.006
< 11,000	3	$\frac{3}{180}$	0.017
< 15,500	14	$\frac{14}{180}$	0.078
< 20,000	36	$\frac{36}{180}$	0.200
< 24,500	90	$\frac{90}{180}$	0.500
< 29,000	139	$\frac{139}{180}$	0.772
< 33,500	180	$\frac{180}{180}$	1.000

An illustration in which we have the cumulative frequency and the relative cumulative frequency is called an Ogive:

Figure 2.7: Cumulative Frequency Distribution of Car Sales Values



2.3 Problem Set